

Congressional Redistricting - Command Reference

Quick Commands (run setup_env.bat first)

File Operations:

ls	List files
ll	Long list
cat file	View file
clear	Clear screen
grep	Search (findstr)
..	Up directory
...	Up 2 levels

Git (GitHub Help):

gs	git status
ga .	git add .
gc "msg"	git commit -m
gp	git push
gl	git log (10)
gd	git diff

Project:

```
run --version v1
  Pipeline
test
  Tests
dash --year 2020
  --version v1
  Dashboard
```

Common GitHub Workflows:

Quick Commit:

```
gs
ga .
gc "your message"
gp
```

Check Before Push:

```
gs
gd
ga .
gc "message"
gp
```

View History:

```
gl
gd
gs
```

Pipeline

```
run --version v1
  # All years (2-4h)
run --year 2020 --version v1
  # Single year (~1h)
run --version v1 --reset
  # Fresh run
run --version v1 --skip-states
  # National only
run --workers 12 --version v1
  # Custom workers
```

Tests

```
test          # All (18s)
pytest tests/unit/ -v # Unit
pytest tests/e2e/ -v  # E2E
pytest tests/ -x      # Stop first
pytest tests/ --cov   # Coverage
```

Dashboard

```
dash --year 2020 --version v1
start outputs/index.html
```

Emergency

```
CANCEL.bat # Kill all Python
```

Claude Skills

```
/run-redistricting
/run-tests
/debug-tests
/debug-pipeline
/enhancement-plan
/enhancement-implement
/create-state-map
/create-national-map
/generate-dashboard
/data-validate
```

Common Flags

Pipeline:

- `--version v1` - Output version
- `--year {2020|2010|2000|all}`
- `--workers N` - Parallel (def: 12)
- `--dpi N` - Image quality (def: 150)
- `--reset` - Delete outputs
- `--skip-states` - National only
- `--print-only` - Dry run
- `--force` - Override skip

Tests:

- `-v` - Verbose
- `-s` - Show prints
- `-x` - Stop on first fail
- `--cov` - Coverage report
- `-k PATTERN` - Filter tests

DPI & Performance

DPI: 100 (dev) — **150 (default)** — 200 (high) — 300 (print)

Times:

- Multi-year: **2-4h**
- Single year: **1h**
- Small state: **30s-2min**
- Large state: **3-5min**
- Tests: **18s**

Key Paths

```
data/tracts/{year}/
  {state}_tracts_{year}.parquet
data/adjacency/{year}/
  {state}_adjacency_{year}.pkl
outputs/us_{year}_{version}/
  states/{state}/
    data/, maps/, political/,
    demographic/, compactness/
```